

Question_9:

Code:

```
1  /*
2  Question_9:
3  Pick exactly P leaves from the stacks to maximize beauty.
4  If a leaf is picked, all leaves above it are also picked.
5  */
6
7  #include <iostream>
8  #include <vector>
9  #include <algorithm>
10 using namespace std;
11
12 int main() {
13     int T;
14     cin >> T;
15
16     while (T--) {
17         int n, k, p;
18         cin >> n >> k >> p;
19
20         vector<int> dp(p + 1, 0);
21         for (int i = 0; i < n; i++) {
22             vector<int> a(k + 1, 0);
23             for (int j = 1; j <= k; j++) {
24                 cin >> a[j];
25                 a[j] += a[j - 1];
26             }
27             for (int j = p; j >= 0; j--)
28                 for (int x = 1; x <= k && x <= j; x++)
29                     dp[j] = max(dp[j], dp[j - x] + a[x]);
30         }
31         cout << dp[p] << endl;
32     }
33     return 0;
34 }
```

Output:

Test Case_1:

```
2
2 4 5
10 10 100 30
80 50 10 50
250
3 2 3
80 80
20 20
15 50
180
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
2
2 4 5
20 20 70 30
60 30 10 50
200
3 2 3
60 70
15 40
20 10
150
PS C:\Users\Prabin Yadav\
```